

Q1 - 24 June - Shift 1

A baseband signal of 3.5 MHz frequency is modulated with a carrier signal of 3.5 GHz frequency using amplitude modulation method. What should be the minimum size of antenna required to transmit the modulated signal ?

- (A) 42.8 m (B) 42.8 mm
(C) 21.4 mm (D) 21.4 m

Space for your notes:

Q2 - 24 June - Shift 2

An antenna is placed in a dielectric medium of dielectric constant 6.25. If the maximum size of that antenna is 5.0 mm. it can radiate a signal of minimum frequency of _____ GHz.

(Given $\mu_r = 1$ for dielectric medium)

Space for your notes:

Q3 - 25 June - Shift 1

Space for your notes:

Q4 - 25 June - Shift 2

Match List I with List II

Space for your notes:

List -I		List – II	
A	Facsimile	I.	Static Document Image
B.	Guided media Channel	II.	Local Broadcast Radio
C.	Frequency Modulation	III.	Rectangular wave
D.	Digital Signal	IV.	Optical Fiber

Choose the correct answer from the following options :

(A) A –IV, B-III, C-II, D-I

(B) A-I, B-IV, C-II, D-III

(C) A –IV, B-II, C-III, D-I

(D) A-I, B-II, C-III, D-IV

Q5 - 26 June - Shift 1

Questions

MathonGo

Choose the correct statement for amplitude modulation:

Space for your notes:

- (A) Amplitude of modulating is varied in accordance with the information signal.
- (B) Amplitude of modulated is varied in accordance with the information signal.
- (C) Amplitude of carrier signal is varied in accordance with the information signal.
- (D) Amplitude of modulated is varied in accordance with the modulating signal.

Q6 - 26 June - Shift 2

A sinusoidal wave $y(t) = 40\sin(10 \times 10^6 \pi t)$ is amplitude modulated by another sinusoidal wave $x(t) = 20\sin(1000\pi t)$. The amplitude of minimum frequency component of modulated signal is :

Space for your notes:

- (A) 0.5 (B) 0.25 (C) 20 (D) 10

Q7 - 27 June - Shift 1

#MathBoleTohMathonGo

Questions

MathonGo

The height of a transmitting antenna at the top of a tower is 25 m and that of receiving antenna is, 49 m. The maximum distance between them, for satisfactory communication in LOS (Line-Of-Sight) is $K\sqrt{5} \times 10^2 \text{ m}$. The value of K is _____.

[Assume radius of Earth is $64 \times 10^5 \text{ m}$] (Calculate upto nearest integer value)

*Space for your notes:***Q8 - 27 June - Shift 2**

We do not transmit low frequency signal to long distances because

- (a) The size of the antenna should be comparable to signal wavelength which is unreal solution for a signal of longer wavelength.
- (b) Effective power radiated by a long wavelength baseband signal would be high.
- (c) We want to avoid mixing up signals transmitted by different transmitter simultaneously.
- (d) Low frequency signal can be sent to long distances by superimposing with a high frequency wave as well.

Therefore, the most suitable options will be :

- (A) All statements are true
- (B) (a), (b) and (c) are true only
- (C) (a), (c) and (d) are true only
- (D) (b), (c) and (d) are true only

Space for your notes:

#MathBoleTohMathonGo

Q9 - 28 June - Shift 1

Match List-I with List-II

	List-I		List-II
A	Television signal	I	03 KHz
B	Radio signal	II	20 KHz
C	High Quality Music	III	02 MHz
D	Human speech	IV	06 MHz

Space for your notes:

Choose the correct answer from the options given below :

- (A) A-I, B-II, C-III, D-IV
- (B) A-IV, B-III, C-I, D-II
- (C) A-IV, B-III, C-II, D-I
- (D) A-I, B-II, C-IV, D-III

Q10 - 28 June - Shift 2

Amplitude modulated wave is represented by

$$V_{AM} = 10 \left[1 + 0.4 \cos(2\pi \times 10^4 t) \right] \cos(2\pi \times 10^7 t).$$

Space for your notes:

The total bandwidth of the amplitude modulated wave is :

- (A) 10 kHz
- (B) 20 MHz
- (C) 20 kHz
- (D) 10 MHz

Q11 - 29 June - Shift 1

Questions

MathonGo

Only 2% of the optical source frequency is the available channel bandwidth for an optical communicating system operating at 1000 nm. If an audio signal requires a bandwidth of 8 kHz, how many channels can be accommodated for transmission :

- (A) 375×10^7 (B) 75×10^7
(C) 375×10^8 (D) 75×10^9

*Space for your notes:***Q12 - 29 June - Shift 2**

The TV transmission tower at a particular station has a height of 125 m. For doubling the coverage of its range, the height of the tower should be increased by :

- (A) 125 m (B) 250 m
(C) 375 (D) 500 m

Space for your notes:

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Answer Key

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q1 (C) Q2 (6) Q3 (B) Q4 (B)
// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q5 (C) Q6 (D) Q7 (192) Q8 (C)
// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q9 (C) Q10 (C) Q11 (B) Q12 (C)
// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

#MathBoleTohMathonGo

Q1 (C)

$$f_c = 3.5 \text{ GHz} \quad f_m = 3.5 \text{ MHz}$$

Side band frequencies are $f_c - f_m$ & $f_c + f_m$, which

are almost f_c

$$\lambda = \frac{c}{f_c}$$

Minimum length of antenna =

$$\frac{c}{f_c} = \frac{\lambda}{4} = \frac{3 \times 10^8}{3.5 \times 10^9 \times 4}$$

$$= 21.4 \text{ mm}$$

Q2 (6)

$$C' = \frac{C}{\sqrt{\mu_r \epsilon_r}} = \frac{3 \times 10^8}{\sqrt{6.25}} = \frac{3 \times 10^8}{2.5}$$

$$f\lambda = 1.25 \times 10^8 \text{ s}$$

$$\Rightarrow f(5 \times 10^{-3} \times 4) = 1.25 \times 10^8$$

$$f = 6.25 \text{ GHz}$$

So $f \approx 6$

Q3 (B)

Very small change in minority charge carriers produces high value of reverse bias current.

Q4 (B)

Question based on the theory given in NCERT.

Q5 (C)

Hints and Solutions

MathonGo

In amplitude modulation, the amplitude of high frequency carrier wave is varied in accordance with message signal

Q6 (D)

$$y(t) = 40 \sin(10 \times 10^6 \pi t)$$

$$x(t) = 20 \sin(1000 \pi t)$$

$$\Rightarrow \omega_c = 10^7 \pi$$

$$\omega_m = 10^3 \pi$$

$$A_c = 40$$

$$A_m = 20$$

Equation of modulated wave = $(A_c + A_m \sin \omega_m t) \sin \omega_c t$

$\sin \omega_c t$

$$= A_c \left(1 + \frac{A_m}{A_c} \sin \omega_m t \right) \sin \omega_c t$$

$$= A_c (1 + \mu \sin \omega_m t) \sin \omega_c t, \quad \mu = \frac{A_m}{A_c}$$

$$= A_c \sin \omega_c t + \frac{\mu A_c}{2} [\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t]$$

Amplitude of minimum frequency =

$$\frac{\mu A_c}{2} = \frac{A_m}{A_c} \times \frac{A_c}{2} = \frac{A_m}{2} = 10$$

Q7 (192)

#MathBoleTohMathonGo

$$\begin{aligned}
 LOS &= \sqrt{2Rh_T} + \sqrt{2Rh_R} \\
 &= \sqrt{2R} \left(\sqrt{h_T} + \sqrt{h_R} \right) \\
 &= \sqrt{2 \times 64 \times 10^5} \left(\sqrt{25} + \sqrt{49} \right) \\
 &= 192\sqrt{5} \times 10^2 \text{ m.} \\
 K &= 192
 \end{aligned}$$

Q8 (C)

- (a) For low frequency or high wavelength size of antenna required is high.
- (b) E P R is low for longer wavelength.
- (c) yes we want to avoid mixing up signals transmitted by different transmitter simultaneously.
- (d) Low frequency signals sent to long distance by superimposing with high frequency.

Q9 (C)

Theory

Q10 (C)

$$\begin{aligned}
 \text{Bandwidth} &= 2 f_m \\
 &= 2 \times 10^4 \text{ Hz} = 20 \times 10^3 \text{ Hz} \\
 &= 20 \text{ kHz}
 \end{aligned}$$

Q11 (B)

Hints and Solutions

MathonGo

$$\text{Frequency at } 1000 \text{ nm} = \frac{3 \times 10^8}{1000 \times 10^{-9}} \Rightarrow 3 \times 10^{14} \text{ Hz}$$

available for channel band width

$$= \frac{2}{100} \times 3 \times 10^{14} \Rightarrow 6 \times 10^{12} \text{ Hz}$$

Bandwidth for 1 channel = 8000 Hz

 $\therefore$ No. of channel

$$= \frac{6 \times 10^{12}}{8 \times 10^3} \Rightarrow \frac{600}{8} \times 10^7 = 75 \times 10^7$$

Q12 (C)

$$\text{Range } d = \sqrt{2Rh}$$

$$d_2 = 2d_1$$

$$\sqrt{2Rh_2} = 2\sqrt{2Rh_1}$$

$$h_2 = 4h_1 = 500 \text{ m}$$

$$\Delta h = 500 \text{ m} - 125 \text{ m} = 375 \text{ m}$$

#MathBoleTohMathonGo